NON-CONVENTIONAL ENERGY SOURCES

Subject Code: 23A03703a | UNIT – III : SOLAR STORAGE & APPLICATIONS, WIND ENERGY

1. Need for Energy Storage

- Solar energy is **intermittent** (night, clouds, seasons) while demand continues → storage matches supply and demand.
- Storage types: **thermal, electrical (battery), chemical (hydrogen), mechanical (pumped hydro, flywheel, compressed air).**
- **Desirable features:** high energy density, low cost, long life, low loss, easy charge/ discharge, safe.

2. Sensible Heat Storage

> Heat stored by **raising the temperature** of a medium without change of phase.

$$Q = m\,C_p\,(T_f - T_i) = \rho\,V\,C_p\,\Delta T$$

- **Liquid media:** water (C_p = 4.18 kJ/kgK – cheap, best up to 100 °C), oil (to 300 °C), molten salts (to 565 °C), liquid sodium.
- **Solid media:** rock bed/ pebble bed (air collectors, $C_p \approx 0.85$ kJ/kgK), concrete, bricks, cast iron.
- Storage capacity depends on **mass, specific heat and ΔT**; water tank is the most common – **insulate** to cut loss.
- **Advantages:** simple, cheap, no phase issues. **Disadvantages:** large volume, temperature falls as heat is withdrawn, heat losses.

❖Rock-bed storage

- Air flows through packed bed of 2–5 cm pebbles; **charge** hot air top → bottom, **discharge** reverse (to keep stratification); used with solar air heaters for crop dryers and space heating.

3. Latent Heat Storage (PCM)

> Heat stored/ released during **phase change** (melting–solidification) at nearly **constant temperature**.

$$Q = m\,[\,C_{sp}(T_m - T_i) + a_m\, h_{sf} + C_{lp}(T_f - T_m)\,]$$

- (solid sensible + latent of fusion + liquid sensible); a_m = fraction melted, h_{sf} = latent heat.
- **Phase change materials (PCMs):**

PCM	Melting point °C	Latent heat kJ/kg
Paraffin wax	50 – 60	200 – 250
Glauber salt $Na_2SO_4{\cdot}10H_2O$	32	250
Calcium chloride hexahydrate	29	190
Fatty acids / stearic acid	55 – 70	180
$NaNO_3$ / KNO_3 salts	220 – 300	100 – 170

- **Advantages:** 5–14 times more energy per unit volume than sensible storage; nearly constant T delivery.
- **Problems:** low thermal conductivity (add fins/ graphite), **supercooling**, phase segregation/ incongruent melting (hydrated salts), corrosion, cost. Encapsulation in tubes/ spheres/ macro-capsules.
- **Uses:** solar water heaters, cooling load shifting, **building walls**, cooking at night.

4. Stratified Storage

- In a vertical **water tank**, hot water (low density) naturally stays at top and cold at bottom with a thin **thermocline** between them.
- Benefits: collector gets **cold water** at the bottom (higher efficiency); consumer gets **hot water** from the top; delivery temperature stays high.
- Keep inlet velocity low (diffusers, baffles) to avoid mixing; tall slim tank (H/D $\geq$ 2–3).
- **Fully mixed vs stratified:** stratified gives ~ 10–20% higher system performance.

5. Thermo-chemical & other storage (brief)

- **Chemical storage:** reversible reactions, e.g. $CaO + H_2O \rightleftharpoons Ca(OH)_2$ + heat; ammoniates; metal hydrides; **solar hydrogen** by electrolysis.
- **Electrical:** lead-acid, Li-ion, flow batteries for PV.
- **Mechanical:** pumped hydro, flywheel, compressed air.
- **Comparison:** Sensible (cheap, bulky) < Latent (compact, constant T) < Chemical (very compact, long-term, complex).

6. Solar Ponds

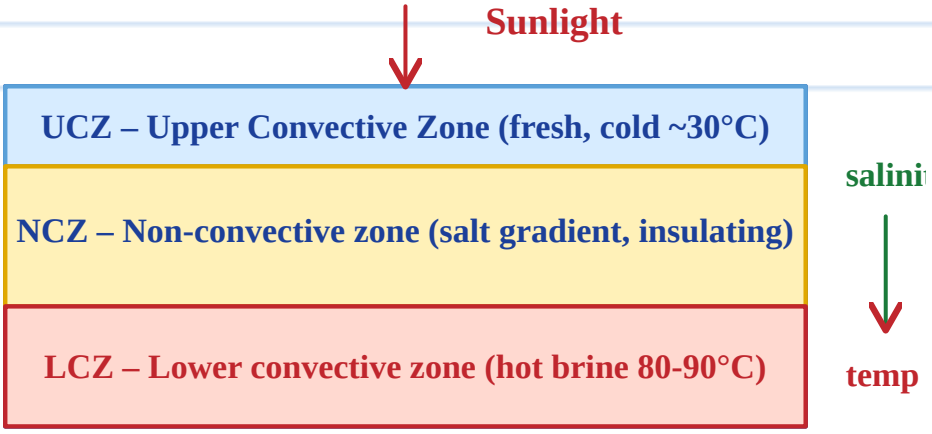


Fig 3.1 Salt-gradient solar pond

> A solar pond is a shallow body of salty water that **collects and stores solar energy** as heat at the bottom without any collector.

❖Three zones

- **UCZ (upper convective zone):** thin (0.1–0.5 m), fresh/ low salinity, near ambient temp, wind-mixed.
- **NCZ (non-convective / gradient zone):** 1–1.5 m, salinity increases with depth → **density gradient suppresses convection;** acts as a **transparent insulator.**
- **LCZ (lower convective zone / storage zone):** 1–2 m, **saturated brine,** hot (70–90 °C), well-mixed, black bottom lining (butyl/ HDPE).

❖Working principle

- **Normal water:** heated water rises → loses heat at top. In a solar pond heated bottom brine is **still denser** than the upper layer → stays at bottom; heat can't escape by convection; only conducts (water poor conductor).

- Heat extracted by pumping hot brine through an external heat exchanger (or an organic Rankine engine).

❖Stability criterion

> $\partial\rho/\partial z > 0$ (density increases with depth) – also $d\rho/dz$ must be sufficient to overcome the thermal expansion

- Maintenance: flush UCZ with fresh water, add salt (NaCl, $MgCl_2$) to LCZ to restore gradient, keep water clear (algae control), control evaporation.

❖Efficiency & uses

- Efficiency of heat collection ~ 15–25%; very low cost per kWh_{th}; Bhuj (Gujarat) 6000 m^2 pond.
- Uses: process heat, space heating, desalination, drying, power (ORC 25–50 kW), aquaculture.
- Others: shallow solar pond, membrane stratified pond.

7. Solar Heating & Cooling Techniques

❖A. Space heating

- Active: collectors + fluid loop + storage + radiators/ fan-coil.
- Passive (no mechanical devices):
  - Direct gain – large south windows; thermal mass floor stores heat.
  - Trombe wall – a massive dark wall behind glass with vents; air circulates by thermosiphon.
  - Sunspace / greenhouse, roof pond, isolated gain.
- Orientation, insulation, thermal mass and overhangs decide performance.

❖B. Solar cooling (refrigeration & air conditioning)

- Vapour absorption cycle ($LiBr–H_2O$ or $NH_3–H_2O$): the generator is heated by solar hot water (80–120 °C); the compressor of conventional system is replaced by absorber + pump + generator → no CFCs, silent.
  - COP ≈ 0.6–0.7 (single effect), 1.2 (double effect).
- Adsorption cooling: silica gel–water or zeolite–water.
- Desiccant cooling: solar regenerates the desiccant wheel; evaporative cooling follows.

- Solar-PV driven compressor chillers, solar ejector cooling, evaporative cooling.
- Passive cooling: wind towers, shading, roof ponds, night radiative cooling, earth tubes.

8. Solar Distillation

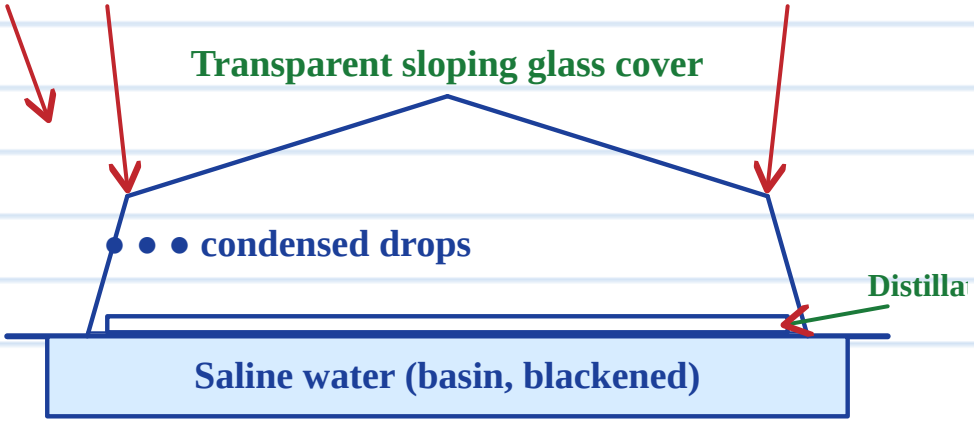


Fig 3.2 Basin type solar still

> Process of converting **saline/brackish water into potable water** using the sun's heat – evaporation followed by condensation.

- **Basin type solar still:** blackened basin with 1–5 cm saline water; sloping glass cover (tilt 10–30°); solar radiation passes the glass, heats water → evaporates; vapour condenses on cool inner glass surface, runs down to the distillate channel.
- **Productivity** ≈ 2–5 L/m²day (≈ 4 L average). Efficiency 30–40%.

$$\text{Hourly distillate } m_d = h_{ew} (T_w - T_g) A / L \text{ [kg/h]}$$

- h_{ew} = evaporative heat transfer coefficient; L = latent heat ≈ 2260 kJ/kg. Efficiency $\eta = m_d L / (A I)$.
- **Improve output:** reduce water depth, insulate basin, add wicks, reflectors, preheat feed, **multi-effect/ double basin**, use dye, cool the glass.
- Types: single-slope, double-slope, tubular, wick, multi-effect, still + FPC (active).
- Uses: drinking water for coastal/ arid villages, battery water, labs.

9. Solar Drying

- Moisture removal from agricultural products (grains, fruits, chilli, fish) using solar heated air.

- **Open sun drying:** cheap, but dust, birds, rain, loss and poor quality.
- **Solar dryers:**
  - **Direct (cabinet) dryer:** product in a glazed box, sun heats the product directly.
  - **Indirect dryer:** air heated in a solar air heater, passes through trays in a drying chamber (colour retained).
  - **Mixed mode** and **hybrid** dryers (with biomass/ electric back-up); **natural vs forced convection**; greenhouse dryer.
- **Advantages:** faster, hygienic, better colour/ quality, less loss, saves fuel.

$$\text{Moisture removed} = W_i - W_f = W_i\,(M_i - M_f)\,/\,(100 - M_f)$$

- Required energy = $m_w L$; air flow rate = $Q / (C_p \Delta T)$.

10. Photovoltaic Energy Conversion

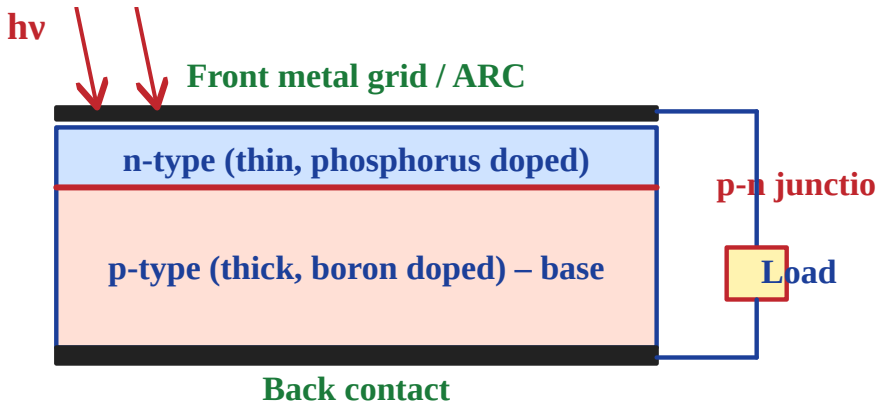


Fig 3.3 Silicon solar cell (photovoltaic effect at p-n junction)

> **Photovoltaic effect:** generation of voltage/ current when light falls on a p-n junction; discovered by Becquerel (1839).

❖Principle

- Photon energy $h\nu \geq$ band gap E_g (Si = 1.1 eV, GaAs 1.4, CdTe 1.5) creates electron–hole pairs.
- **Built-in electric field** at the junction (depletion region) separates them: electrons → n side, holes → p side → current through external load.
- Photons with $h\nu < E_g$ are not absorbed; with $h\nu > E_g$ excess is lost as heat → theoretical Si limit ~ 29–30% (Shockley–Queisser).

❖Cell characteristics

- I–V relation (ideal): $I = I_L - I_o\,[\exp(qV / nkT) - 1]$

- Short-circuit current $I_{sc} \approx I_L$; Open-circuit voltage $V_{oc} = (nkT/q) \ln(1 + I_L/I_o) \approx 0.6$ V (Si).
- Maximum power point (V_m, I_m); Fill factor $FF = V_m I_m / (V_{oc} I_{sc}) \approx 0.7–0.8$

$$\text{Efficiency } \eta = P_{max} / (I \times A) = FF \cdot V_{oc} \cdot I_{sc} / (I \times A)$$

- Standard Test Conditions (STC): 1000 W/m², 25 °C cell temp, AM 1.5.
- Temperature effect: V_{oc} falls ~ 2.3 mV/°C/cell; efficiency falls ~0.4–0.5%/°C.

❖Types of cells

Type	Efficiency
Mono-crystalline Si	18 – 23 %
Poly-crystalline Si	15 – 18 %
Thin film (a-Si, CdTe, CIGS)	8 – 18 %
Multi-junction (GaAs)	30 – 45 %
Perovskite / organic	15 – 25 % (research)

❖From cell to system

- 1 cell ≈ 0.5-0.6 V; 36 cells in series → 12 V module; modules in series (↑ V), parallel (↑ I) form array.
- System parts: PV array, charge controller (MPPT), battery, inverter (DC → AC), mounting, tracker; grid-connected (net metering) or stand-alone.
- Uses: street lights, water pumping, rooftops, satellites, solar parks, calculators, telecom towers.

II. WIND ENERGY – Sources & Potential

- Wind = air in motion, caused by uneven solar heating of earth's surface (≈ 1–2% of solar energy converts to wind), Coriolis force, and topography.
- Global patterns: trade winds, westerlies, polar easterlies, monsoons, land & sea breezes, mountain-valley winds.
- Wind power potential: world ~ 10^6 GW theoretical; India installed ≈ 50 GW+ (4th in the world); best sites need mean speed > 5–6 m/s (Tamil Nadu – Muppandal, Gujarat, Karnataka, Rajasthan, A.P. – Ramagiri, Maharashtra).

- **Advantages:** free, clean, fast installation, land can still be farmed.
  Disadvantages: intermittent, noise, bird/ bat hits, visual impact, site-specific, grid integration.

❖Wind speed variation

- **Height:** power-law $V / V_o = (h / h_o)^{\alpha}$, $\alpha \approx 1/7$ (0.14) open land, 0.25–0.4 urban/forest.
- **Weibull distribution** describes frequency of wind speeds: $f(v) = (k/c)(v/c)^{k-1} \exp[-(v/c)^k]$.
- Wind measured by **cup anemometer**, wind vane; Beaufort scale.
- **Wind rose** diagram: direction frequency.

12. Power in the Wind

- Kinetic energy of air of mass m moving at velocity V: $KE = \frac{1}{2} m V^2$.
- Mass flow through rotor area A: $\dot{m} = \rho A V$.

$$P_{wind} = \frac{1}{2} \rho A V^3 = \frac{1}{2} \rho (\pi D^2/4) V^3 \ [W]$$

- $\rho = 1.225\ kg/m^3$ (sea level, 15 °C). Power $\propto V^3$ and $\propto D^2$ – doubling wind speed gives 8 × power!
- **Power density** = $\frac{1}{2} \rho V^3\ W/m^2$ (e.g. 10 m/s → 612 W/m^2).
- Extracted power $P = C_p \cdot \frac{1}{2} \rho A V^3 \cdot \eta_{mech} \eta_{gen}$.
- Tip-speed ratio $\lambda = \omega R / V = \pi D N / (60 V)$.
- **Solidity** = blade area / swept area.

13. Horizontal Axis Wind Turbine (HAWT)

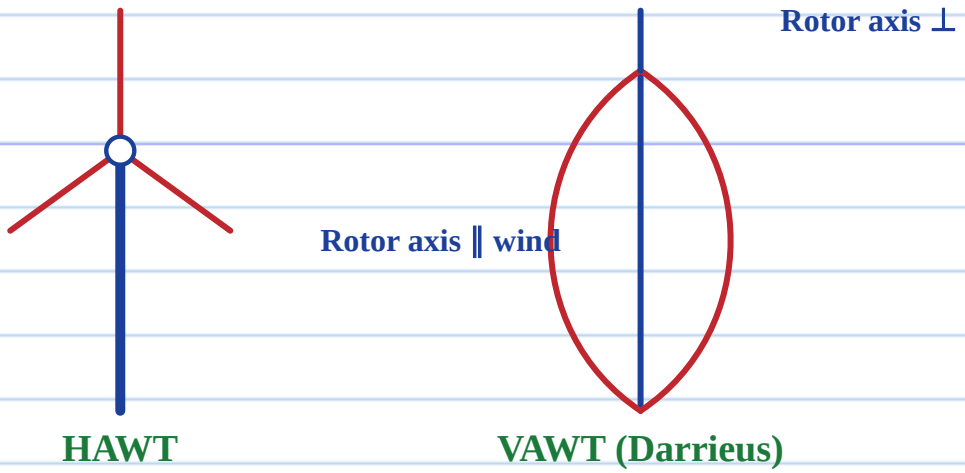


Fig 3.4 Horizontal axis (left) and vertical axis (right) wind turbines

- Rotor shaft **parallel to wind**; main components: **rotor blades** (2–3), **hub**, **nacelle** (gearbox, generator, brake, controller), **tower**, yaw mechanism to face the wind, pitch control.
- **Types:** 3-blade upwind (modern), 2-blade, down-wind, multi-blade (American farm windmill – water pumping), Dutch type windmill.
- **Advantages:** high C_p (0.40–0.50), self-starting (multi-blade), tall tower reaches steadier faster wind, pitch control.
- **Disadvantages:** needs yaw system; heavy gearbox/ generator at top; costly tower & maintenance; noise.
- **Rated wind speed** 10–14 m/s; **cut-in** 3–4 m/s; **cut-out** 20–25 m/s (feathered/ braked).

❖**Typical sizes**

- Small (< 10 kW) – farms; medium (100 kW–1 MW); large 2–15 MW (offshore).

14. Vertical Axis Wind Turbine (VAWT)

- Rotor shaft **perpendicular to wind** – accepts wind from any direction (no yaw).
- **Types:**
  - **Savonius rotor** – S-shaped drag type; self-starting, low speed, $C_p \approx 0.15$–0.3 – used for water pumping.
  - **Darrieus rotor** (eggbeater) – lift type, curved blades; high speed, $C_p \approx 0.35$; **not self-starting.**
  - H-Darrieus / Giromill – straight blades with pitch control.
- **Advantages:** generator/ gear at ground; no yaw; simple; works in turbulent wind.
- **Disadvantages:** lower efficiency, needs starting, blades see low wind close to ground, fatigue issues.

❖HAWT vs VAWT

Feature	HAWT	VAWT
Axis	Horizontal	Vertical
Yaw control	Needed	Not needed
Efficiency (C_p)	0.40–0.50	0.15–0.35
Self-starting	Yes (multi-blade)	No (Darrieus)
Generator location	Top of tower	Ground
Common use	Power plants	Small/ urban, pumping

15. Betz Criterion (Betz Limit)

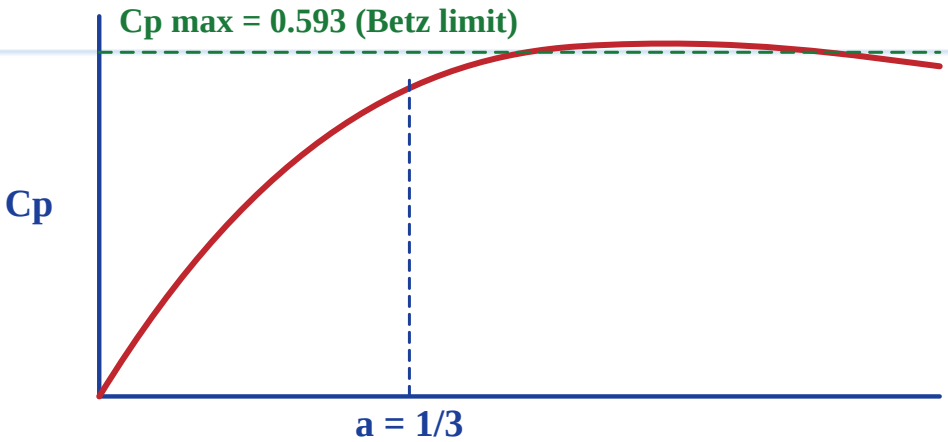


Fig 3.5 Power coefficient vs. induction factor (peak at a = 1/3)

> Betz limit: no wind turbine can convert more than 16/27 = 59.3 % of the kinetic energy of the wind into mechanical energy.

❖Derivation (momentum theory – ideal rotor)

- Wind speeds: V_1 (upstream, far), $V = (V_1 + V_2)/2$ at rotor (actuator disc), V_2 (downstream, far).
- Mass flow $\dot{m} = \rho A V = \rho A (V_1 + V_2)/2$.
- Power extracted = change in KE per sec: $P = \frac{1}{2} \dot{m} (V_1^2 - V_2^2) = \frac{1}{4} \rho A (V_1 + V_2)(V_1^2 - V_2^2)$
- Divide by available power $P_o = \frac{1}{2} \rho A V_1^3$. Let $x = V_2 / V_1$:

$$C_p = P / P_o = \frac{1}{2} (1 + x)(1 - x^2)$$

- Maximise: $dC_p/dx = 0 \rightarrow \frac{1}{2} [(1 - x^2) + (1 + x)(-2x)] = 0 \rightarrow 1 - 2x - 3x^2 = 0 \rightarrow x = 1/3$.

$$C_{p,max} = \frac{1}{2} (4/3)(1 - 1/9) = \frac{1}{2} \times 4/3 \times 8/9 = 16/27 = 0.593$$

- So at the optimum, $V = (V_1 + V_1/3)/2 = 2V_1/3$ and wake speed $V_2 = V_1/3$.
- In terms of axial induction a: $V = V_1(1 - a)$, $C_p = 4a(1 - a)^2$, max at $a = 1/3$.
- Real turbines reach $C_p = 0.35–0.50$ because of drag, wake rotation, tip losses, finite blades.
- Overall efficiency = $C_p \times \eta_{gear} \times \eta_{gen}$.

16. Performance Characteristics

- Power curve: P vs wind speed – zero below cut-in, rises as V^3 to rated speed, flat at rated power (pitch/ stall control), shuts at cut-out.
- $C_p - \lambda$ curve: C_p rises with tip-speed ratio to peak (λ_{opt} = 6–8 for 3-blade HAWT, 4–5 multi-blade slower, 2–3 Savonius), then falls.
- Torque coefficient $C_T = C_p/\lambda$; high at low λ for drag machines (good starting).
- Capacity factor (CUF) = actual energy / (rated power × 8760 h) ≈ 20–35% on land, 40–50% offshore.
- Power control: pitch control (blades rotate), stall control, yaw control, brake.
- Wind farm layout: spacing 5–9 D downwind, 3–5 D crosswind to limit wake losses.

17. Solved Problems

❖Problem 1

Wind at 10 m/s, rotor diameter 40 m, $\rho = 1.2$. $A = \pi(20)^2 = 1256.6\ m^2$. $P_{wind} = 0.5 \times 1.2 \times 1256.6 \times 1000 = 754$ kW. With $C_p = 0.4 \rightarrow P = 301.6$ kW; Betz max = 0.593 × 754 = 447 kW.

❖Problem 2 – Tip speed ratio

D = 40 m, N = 30 rpm, V = 10 m/s: $\lambda = \pi D N/(60 V) = \pi \times 40 \times 30/(600) = 6.28$.

❖Problem 3 – Wind speed at height

V at 10 m = 5 m/s, $\alpha = 1/7$. At 80 m: $V = 5 \times 8^{1/7} = 5 \times 1.346 = 6.73$ m/s; power rises by $(1.346)^3 = 2.44$ times.

❖Problem 4 – Annual energy

> 2 MW turbine, CUF = 30 % → $E = 2000 \times 8760 \times 0.30 = 5.26 \times 10^6$ kWh/yr.

18. Important Questions

- Why energy storage? Explain sensible, latent heat and stratified storage.
- Explain working of a salt-gradient solar pond with a neat sketch.
- Describe solar distillation (basin still), solar drying, solar absorption cooling.
- Explain the principle of PV conversion; I–V characteristics, fill factor, efficiency.
- Compare HAWT and VAWT; explain Savonius and Darrieus rotors.
- Derive Betz limit and show $C_{p,max} = 0.593$.